What Should Occupy the Context Budget? Latent Compression versus Budget-Matched Truncation in Document QA

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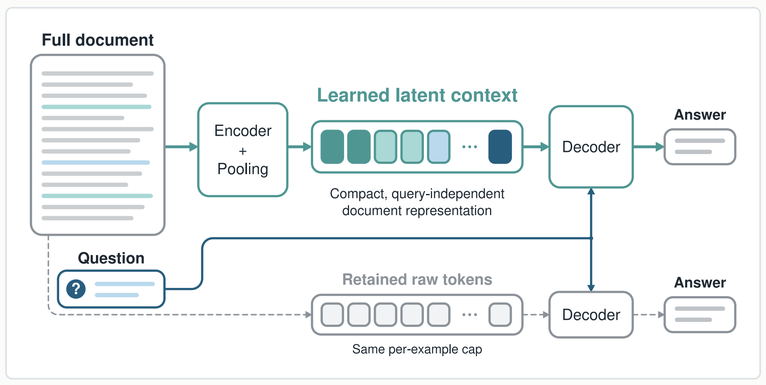
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CODE & EVALUATION ARTIFACTS SCAN →

Given the same per-example decoder-context cap, a minimal compressor (text encoder, learned pooling, direct projection) beats raw-token baselines whose readers were trained on their own inputs, at 8× and 16× compression ratios. Adding transformer depth after the pooling bottleneck makes it worse.

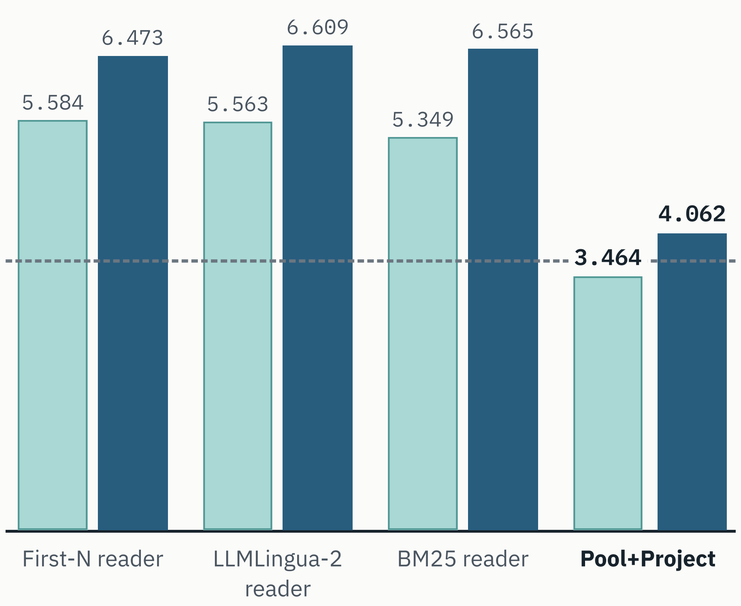
01 ONE FIXED BUDGET, TWO WAYS TO FILL IT

A serving system allocates a fixed number of decoder positions to a document. It can keep raw tokens, or replace the document with learned vectors. We give both the **same cap per example** and compare them.



SAME CAP	Every policy gets V = ⌈L / cr⌉ document positions; sample IDs and caps verified before pairing.
FAIR READERS	First-N, LLMLingua-2 and question-aware BM25 each get a reader trained on their own outputs.
DOC-CLUSTERED	~15 questions share a document, so every 95% CI resamples documents, not rows.

02 COMPRESSION WINS AT A MATCHED BUDGET



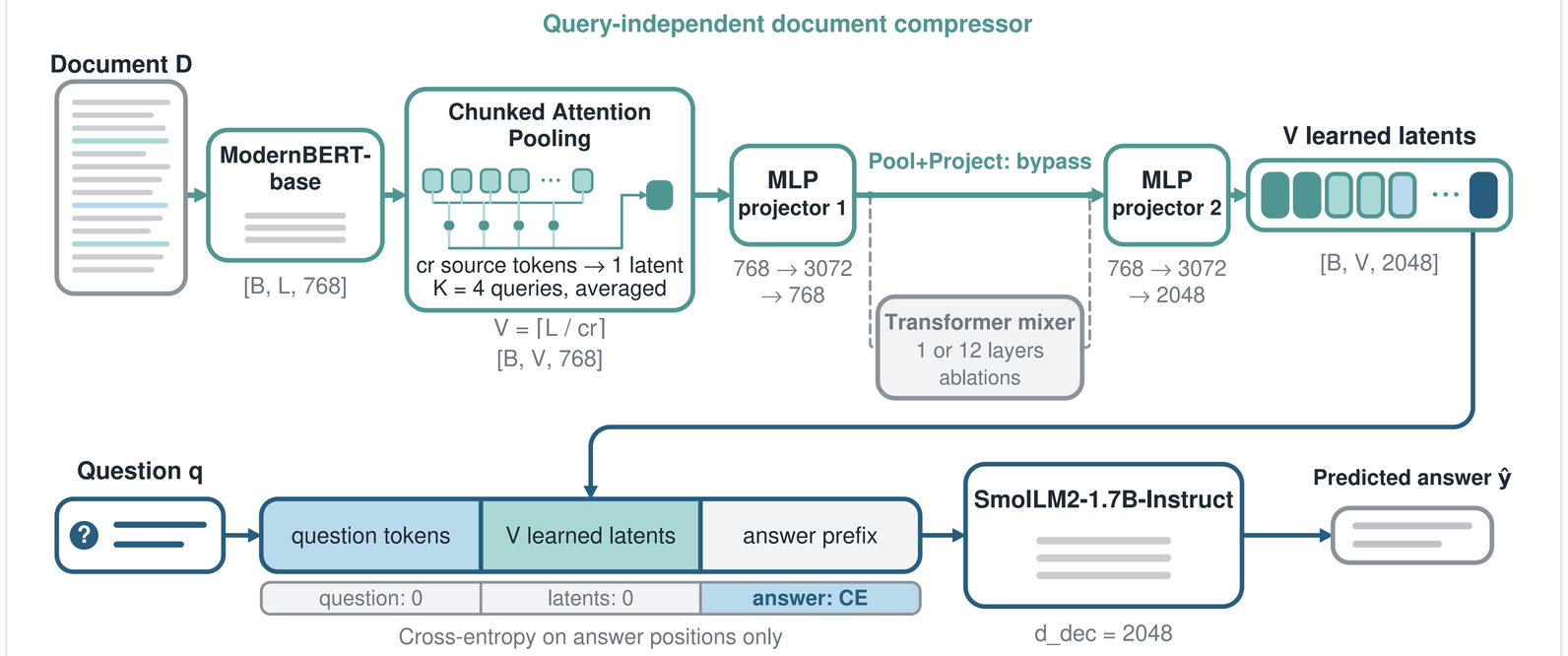
Held-out answer perplexity on FRESH, lower is better.

At the same per-example cap, Pool+Project has the best perplexity and token accuracy at both rates, against readers trained on each selector's own outputs.

READER OF THE CONTEXT	CR = 8 · PPL / TA	CR = 16 · PPL / TA
Pool+Project (ours)	**3.464 / 0.681**	**4.062 / 0.649**
First-N, trained reader	5.584 / 0.610	6.473 / 0.585
LLMLingua-2, trained reader	5.563 / 0.608	6.609 / 0.580
BM25, trained reader	5.349 / 0.620	6.565 / 0.584
Full document, frozen (ref.)	3.653 / 0.719	same

3,000 held-out pairs over 205 documents. Every paired ΔNLL interval excludes zero, e.g. vs BM25 −0.435 [−0.461, −0.409] at cr = 8 and vs LLMLingua-2 −0.487 [−0.512, −0.461] at cr = 16. The frozen full-document reader is a reference without a budget.

03 POOL+PROJECT: ENCODE, POOL, PROJECT



Chunked attention pooling maps every cr encoder tokens to one latent; MLP projectors map it into the decoder after the question. One answer-token cross-entropy loss trains everything end-to-end; the document is never reconstructed. Dashed: the ablated post-pooling mixer (1 or 12 layers).

04 HELD-OUT PROTOCOL

18,146	PwC QA test triples over 1,224 Pile documents (ICAE); 0 train/test document overlap
3,000	FRESH headline pairs, 205 documents, disjoint from the development slice
1→16	Curriculum cr = 1→2→4→8→16, 11,325 AdamW updates; trained end-to-end, only encoder input embeddings frozen

Primary metrics: teacher-forced answer perplexity (PPL) and token accuracy (TA). Generation is secondary because systems stop differently.

05 SAME STORY ON QWEN2.5-0.5B

13.75 → 5.40 PPL at cr = 8, first-N reader → ours

16.96 → 6.92 PPL at cr = 16

Paired ΔNLL −0.935 [−0.985, −0.884] and −0.897 [−0.942, −0.851]; TA +10.2 and +8.7 points. Depth ordering repeats: 6.917 vs 16.820 with the 12-layer mixer.

06 FREE-RUNNING ANSWERS AGREE

+0.030	token-F1, first sentence, cr = 16 (0.369 vs 0.338) · CI [+0.023, +0.038]
+0.064	answer containment · CI [+0.057, +0.072]
46.7%	manual acceptability vs 36.0% on 150 answers, same direction

QWEN3-8B JUDGE VS MATCHED FIRST-N READER · RANDOMIZED A/B, DESCRIPTIVE



07 DEPTH AFTER POOLING HURTS

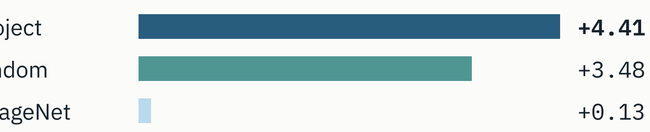
Same model with a 12-layer transformer mixer inserted after pooling. Held-out PPL at cr = 16, document-cluster 95% CI.

RUN	OURS	+12L	ΔPPL [95% CI]
seed 42	**4.062**	4.688	+0.63 [0.57, 0.68]
seed 43	**4.061**	4.715	+0.65 [0.60, 0.70]
seed 44	**4.057**	4.951	+0.89 [0.82, 0.96]

Monotone over depths {0, 1, 12}: even one block costs +0.027 nats [+0.023, +0.031]. ImageNet-ViT initialization widens the gap; freezing widens it again. The penalty persists with fp32 optimizer masters.

08 THE WINNER READS THE DOCUMENT

Randomly permuting source tokens keeps their multiset but destroys order. The strongest system is the most order-sensitive; weaker branches ignore order.



PPL increase when tokens are permuted, cr = 16, FRESH. Zeroing the learned pool queries costs +0.186 nats.

09 INSIDE THE MIXER

EFFECTIVE RANK	**170** bypass · 162 random · 85 ImageNet-init after the block
IN/OUT COSINE	**1.00** bypass · 0.44 random · 0.30 ImageNet: the mixer rotates latents
WEIGHT DRIFT	3.9% for ImageNet weights vs 29.5% for random: the vision prior barely adapts

256 held-out documents, cr = 16, descriptive. Padding is ruled out: toggling the key mask moves PPL ≤ 0.53% against a 15.4% depth gap.

10 SMALLER, FASTER, CACHEABLE

	OURS	+12L	Δ
Params*	**163.2M**	249.6M	**−35%**
Fwd, cr=16	**60.2 ms**	65.0 ms	**−7.4%**
Peak mem	**3.92 GB**	4.09 GB	**−4.2%**

*Non-decoder modules; A100 bf16 forward microbenchmark. Latents are computed once per document (0.125 MB at cr = 16). Amortized over 15 questions, a query costs 85.8 ms, **within 2.3% of raw-token truncation**.

Scope. One English QA corpus with GPT-4-written answers and documents under 1,024 tokens; two decoders up to 1.7B. The frozen full-document reader keeps higher TA (0.719) and wins at cr = 16, so it is reported as a reference, not a baseline.